

Proposal to Create Handles Task Force

Abstract

In this proposal, we are requesting that the Social Web WG form a Handles Taskforce to begin working toward including domains as a first-class handle alternative for ActivityPub, in addition to the existing `@username@WebFinger-domain` pattern. We propose HTTPS well-known as the primary resolution mechanism, but are open to WebFinger and DNS as options as well.

Background

ActivityPub currently uses an `Actor` object with `preferredUsername`. The handle format used across the ecosystem is `@username@WebFinger-domain`.

Another pattern that is becoming popular is bare domain-based handles (ie. `@example.com` instead of `@username@example.com`). There are various benefits to this pattern:

- It removes the username and domain split
- Leverages domain ownership as a trust signal
- Sidesteps some, but not all, impersonation risks

The proposed Handles Taskforce is scoped to cover handles as a technical concept, including resolution mechanisms, internationalization, and Actor verification. This proposal requests that domain-based handles resolved via HTTPS well-known be an additional mechanism formally specified within that scope.

Proposal

Handle Format

A domain-based handle is a fully-qualified domain name used directly as the handle, with no separate username component. Ownership of the domain via DNS delegation is treated as ownership of the handle.

This is not a replacement for the `@username@WebFinger-domain` format, but instead an additional, optional format alongside the existing pattern.

Resolution Mechanism

We propose that the resolution primarily be performed via an HTTPS request to a well-known path on the handles domain.

GET <https://{domain}/.well-known/activitypub-actor>

This should return a JSON document that resolves the domain to its corresponding Actor URI, and potentially other information in the future.

Internationalization

To better support the broader global audience, outside of regions where the Latin character set are dominant, we need to support internationalization. Internationalization is also one of the W3C's Mission items: <https://www.w3.org/mission/internationalization/>

Work in this area will ensure that all parts of the handle for an account (whether `@domain.example` or `@username@domain.example`) both work for international audiences, whilst also preventing common issues such as homoglyph attacks. For this area of work, we'll rely on community members who've had experience with internationalization in other ecosystems, such as email.

Security

The main security consideration for this task forces work is that handles must be bidirectionally verified, that is when you resolve the handle to an Actor ID, you can then request that Actor ID and receive back a property that points to the original handle.

Verification

It's also required that the Actor is verifiable as the legitimate owner of the domain handle. We propose the well-known document include a direct reference to the Actor's URI. A simple, explicit string/URI field is sufficient to verify correctly.

Next Steps

- Create a Handles Taskforce for domain-based handles
- Specify HTTPS well-known resolution document format
- Align on internationalization and security requirements
- Document WebFinger and DNS as alternative resolution mechanisms (?)